

EXPERIMENT 2

MONOALPHABETIC SUBSTITUTION CIPHER (PYTHON)

AIM

To implement the Monoalphabetic Substitution Cipher using Python.

ALGORITHM

1. Read the plaintext from the user.
2. Define a substitution key containing all 26 unique alphabet letters.
3. Replace each plaintext letter with its corresponding letter from the substitution key.
4. Display the encrypted (cipher) text.
5. For decryption, replace each cipher letter with its corresponding original alphabet letter.
6. Display the decrypted text.

CODE

```
import string

alphabet = string.ascii_uppercase
key = "QWERTYUIOPASDFGHJKLZXCVBNM"

plain = input("Enter the plain text: ").upper()

# Encryption
cipher = ""
for ch in plain:
    if ch.isalpha():
        index = alphabet.index(ch)
        cipher += key[index]
    else:
```

```

cipher += ch

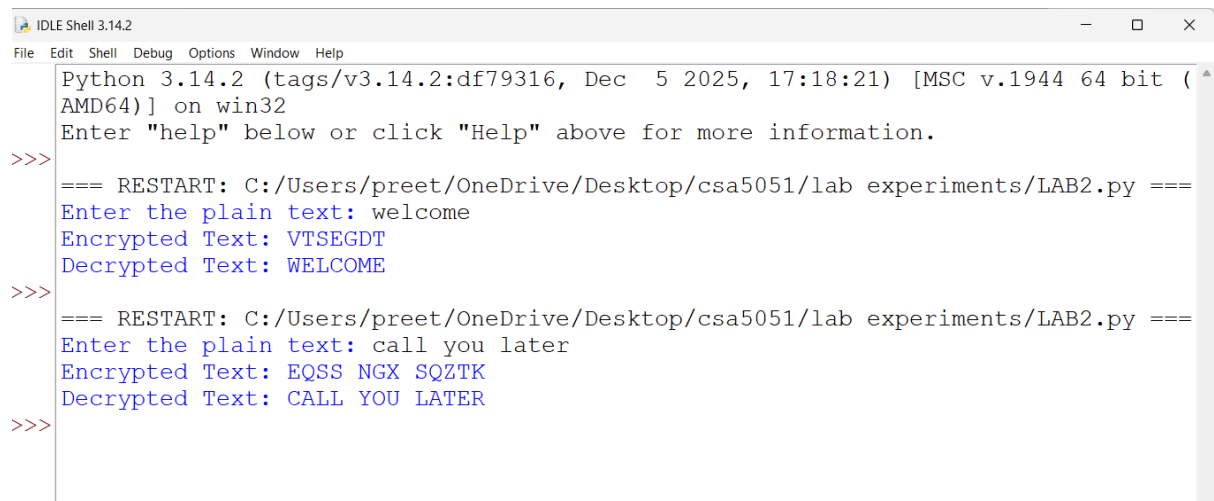
print("Encrypted Text:", cipher)

# Decryption
decrypted = ""
for ch in cipher:
    if ch.isalpha():
        index = key.index(ch)
        decrypted += alphabet[index]
    else:
        decrypted += ch

print("Decrypted Text:", decrypted)

```

OUTPUT



```

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB2.py ===
Enter the plain text: welcome
Encrypted Text: VTSEGDT
Decrypted Text: WELCOME
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB2.py ===
Enter the plain text: call you later
Encrypted Text: EQSS NGX SQZTK
Decrypted Text: CALL YOU LATER
>>>

```

RESULT

Thus, the Monoalphabetic Substitution Cipher was implemented successfully using Python.